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Non-intrusive measurement of magnetic field strengths in a low-pressure argon plasma using quantum beat spectroscopy

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Abstract

Accurate measurement of magnetic field strengths is critical in many plasma environments, ranging from astrophysical systems to fusion energy research. In this work, a non-perturbative laser-based optical diagnostic known as *quantum beat spectroscopy* is demonstrated to be good alternative for measuring the magnetic field strength in low-pressure laboratory plasmas. The technique is investigated using both ns and fs pulsed lasers in an argon plasma. Preliminary results for a helium plasma are also given. Zeeman-split $J = 1$ electron states with transitions from metastable states were identified and tested for neutral argon ($^2P_{1/2}^o$) $4p^2[1/2]$ and neutral helium $1s3p^1P_1^o$. Magnetic fields are measured with sub-Gauss precision at near single laser pulse acquisition rates.

Keywords: optical magnetic field diagnostic, laser spectroscopy, quantum beat spectroscopy, laser induced fluorescence

1. Introduction

Magnetic fields play a crucial role in the dynamics of plasmas. Therefore, diagnostics to accurately and efficiently measure magnetic fields are essential for plasma experiments. For laboratory magnetic reconnection experiments [1–4], methods to non-intrusively monitor the spatial and temporal evolution of weak magnetic and electric fields throughout the magnetic reconnection process provide important insight into the

physical processes that drive magnetic reconnection. Existing methods to measure magnetic fields in plasma environments have their own strengths and weaknesses.

Miniature Hall probes are commercially available and accurate over a wide range of magnetic fields, but are very sensitive to ambient temperature and therefore perform poorly in even moderate temperature plasmas.

Inductive ‘sense coil’ probes are relatively easy and inexpensive to construct but are perturbative and average over many of the scales of interest in reconnection experiments [5]. These probes measure the time-varying magnetic field with a simple loop of wire by measuring the induced voltage on the coil. By integrating the output voltage, the magnetic field strength through the loop cross section is determined. Such probes are also prone to noise generated by ambient electromagnetic fields, are insensitive to DC fields, and have a limited frequency range of operation.

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Magneto-optical probes avoid some of the difficulties of magnet sense coils [6]. Przybysz *et al* showed field fluctuations as small as 1 G were resolvable by measuring small changes in the polarization of laser light passing through a magnetized plasma using a set of fibers and an optical crystal. While the magneto-optical probe does not suffer from the frequency limitations of magnetic sense coils, the experimental apparatus is complex, the probes are still perturbative, and the sensitivity is limited by the Verdet constant of the crystal. Polarimetry is another optical method of measuring magnetic fields in plasma. In polarimetry, the Faraday rotation of laser light passing through plasma is measured [7]. Polarimetry requires high plasma densities and large magnetic field strengths and is line integrated.

In magnetic fusion experiments, the motional Stark effect (MSE) is used to measure the effects that a strong magnetic field has on the emission spectra of a neutral beam [8]. MSE uses a high energy neutral hydrogen beam injected across a strong magnetic field to generate an electric field in the rest frame of the atom. The electric field induces Stark splitting in the observed fluorescence of the neutral beam atoms after excitation from collisions with the background plasma. From the polarization of the neutral fluorescence and the relative transition amplitudes, the magnetic field strength is derived. That MSE requires a strong magnetic field and a high-energy neutral hydrogen beam makes it suitable only for complex magnetic fusion experiments.

Laser induced fluorescence (LIF) of Zeeman-split absorption lines has been employed to measure magnetic fields with moderate precision (~ 10 G) in laboratory plasmas [9, 10]. Deriving the magnetic field from an LIF measurement necessitates capturing the entire Doppler-broadened Zeeman σ -peak line shape by sweeping the laser wavelength over a substantial range. The requirement to raster the laser wavelength increases the acquisition time to measure a single magnetic field magnitude. Doppler-broadening of the line shape also places a lower bound on the minimum magnetic field strengths measurable with this approach (Doppler broadening widens the individual Zeeman split states until they are indistinguishable for weak magnetic fields). The Doppler broadening effects are reduced somewhat by measuring the line shape of each σ -peak separately, but at the cost of increased time to acquire a single magnetic field measurement. The time required prevents measurements of transient magnetic effects.

Here we present measurements of magnetic fields in a magnetized argon plasma using the method of quantum beat spectroscopy (QBS) [11]. QBS is a purely optical, pulsed laser-based diagnostic that provides a Doppler-free, near single-shot measurement of the magnetic field strength. We show that magnetic field strengths as small as 12 G with sub-Gauss resolution are non-intrusively measured using QBS in only one to a few laser pulses. This method for measuring magnetic field strengths is ideal for investigating systems with weak, transient electromagnetic events, i.e. laboratory magnetic reconnection. In section 2, we discuss the theory of QBS and its application as a magnetic field diagnostic. The experimental facility and laser systems used for this work are described in

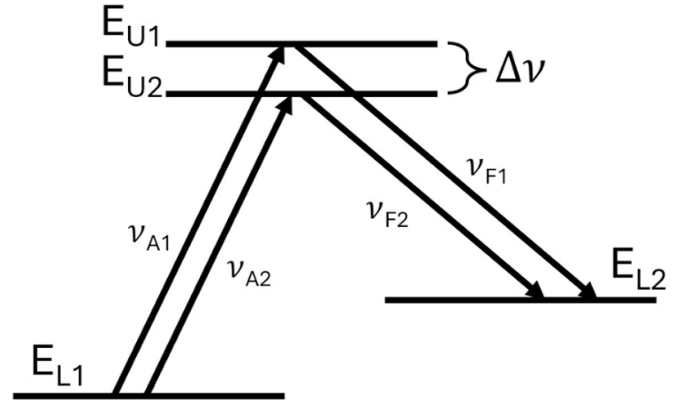


Figure 1. An electron in the lower energy level is excited by photon absorption into a coherent superposition of the upper energy levels. When the electron relaxes to a common lower energy level, the uncertainty of which excited state the electron relaxed from results in an interference pattern observed in the fluorescence as a beat frequency of $\Delta\nu$.

section 3. Lastly, magnetic field strengths derived from QBS for various operating conditions and closing remarks are given in sections 4 and 5, respectively.

2. Quantum beat spectroscopy

QBS is a non-perturbative, laser-based technique used in a variety of physical systems to investigate the separation of two electron states that have an allowed transition to a common lower energy state. This technique has been employed to measure Zeeman splitting [11–14], Stark splitting [15], fine structure [11, 16–18], hyperfine structure [19, 20], and to generate THz radiation [21–23].

Extensive treatments of QBS are available in the literature [24, 25] and the concept is only briefly summarized here. Consider a population of atoms with electrons in a lower energy state $|\psi_{L1}\rangle$ with energy E_{L1} with allowed transitions to closely spaced upper states $|\psi_{U1}\rangle$ and $|\psi_{U2}\rangle$ with energies E_{U1} and E_{U2} , respectively (figure 1). A short-pulse laser with spectral linewidth $\Delta\nu_{\text{las}} > (E_{U2} - E_{U1})/\hbar$ is then able to excite an electron into a coherent superposition of both upper energy states at time $t = 0$:

$$\psi(0) = \sum_k a_k |\psi_k(0)\rangle = a_{U1} |\psi_{U1}(0)\rangle + a_{U2} |\psi_{U2}(0)\rangle, \quad (1)$$

where the probability of the state k is given by $|a_k|^2$. The excited electrons exponentially relax down to a common lower energy level $|\psi_{L2}\rangle$ with the decay constant $\gamma_k = 1/\tau_k$, such that the evolution of the state is given by

$$\psi(t) = \sum_k a_k |\psi_k(0)\rangle e^{-i(E_k - E_{L2})/\hbar - \gamma_k/2)t}. \quad (2)$$

A detector collects the fluorescence from both upper excited states such that the signal is proportional to $|\langle\psi_{L2}|\epsilon \cdot \mu|\psi(t)\rangle|^2$ where ϵ is the polarization direction of

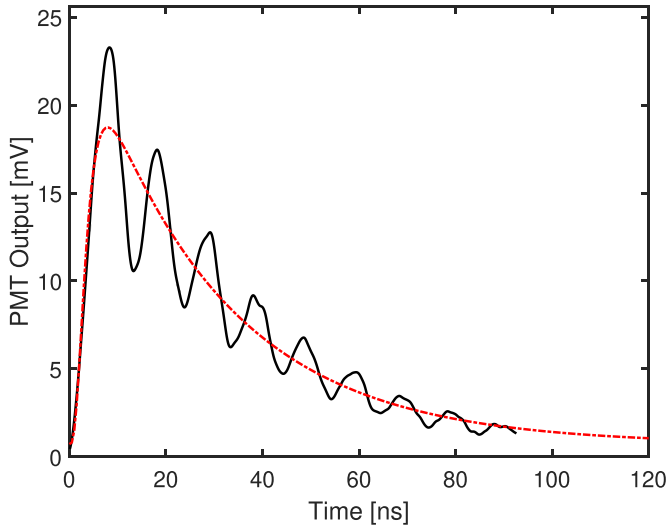


Figure 2. An example photo-multiplier tube (PMT) trace of the measured fluorescent signal due to quantum beat interference (solid black). The standard laser-induced fluorescent signal (dot-dash red) is superimposed on the measured QBS signal.

the fluorescence and μ is the dipole operator of the atom. If we assume a common decay constant, this gives a fluorescence signal of

$$S(t) \propto e^{-\gamma t} \left(a_{U1}^2 |\langle \psi_{L2} | \epsilon \cdot \mu | \psi_{U1}(0) \rangle|^2 + a_{U2}^2 |\langle \psi_{L2} | \epsilon \cdot \mu | \psi_{U2}(0) \rangle|^2 + 2a_{U1}a_{U2} |\langle \psi_{L2} | \epsilon \cdot \mu | \psi_{U1}(0) \rangle| \cdot |\langle \psi_{L2} | \epsilon \cdot \mu | \psi_{U2}(0) \rangle| \cos((E_{U2} - E_{U1})t/\hbar) \right) \quad (3)$$

which takes the form of a superposition of a decaying exponential and a damped oscillator, coined as a ‘quantum beat,’ shown in figure 2. The frequency of the modulation is identical to the difference in the Bohr frequencies of the excited states $\omega = (E_{U2} - E_{U1})/\hbar$ and is referred to as the ‘beat frequency.’ This beating effect is analogous to Young’s double slit experiment, where the uncertainty of which state (or slit) is re-emitting the fluorescent photon results in the total fluorescence being the square of the sum of the probability amplitudes of each possibility, producing an interference pattern. When an electron is excited into a superposition of more than two states, the resultant oscillation is a linear combination of cosines with frequencies equal to the differences in the Bohr frequencies of the respective energy levels. Any attempt to distinguish which state the electron was in (such as through a narrow bandpass filter isolating one of the emission lines) results in the absence of the beating signal [24].

A magnetic field will split an energy level into $2J+1$ energy levels where J is the total angular momentum of the state. The splitting is linearly proportional to the magnetic field strength, given by $\Delta E = \beta m_J g_J B$, where β is the Bohr magneton, m_J is the magnetic quantum number for the given electron energy state, g_J is the Landé-g factor for the state, and B is the magnetic field.

For the measurements presented here, neutral argon is excited from the $J=0$, $(^2P_{1/2}^o)4s^2[1/2]^o$ state to the

$J=1$, $(^2P_{1/2}^o)4p^2[1/2]$ state using 772.6333 nm (vac.) photons. Fluorescence is collected from the same transition. Preliminary measurements were also taken in neutral helium, excited from the $J=0$, $1s2s^1S_0$ state to the $J=1$, $1s3p^1P_1^o$ state using 501.7077 nm (vac.) photons. These transitions were chosen to exploit the Zeeman splitting characteristics of the lower and upper state. In the presence of an externally applied magnetic field, the lower state exhibits no Zeeman splitting since the electronic angular momentum is 0. The upper state will split into three unique states with $m_J = -1, 0$, or 1 . This transition provides a simple Zeeman spectrum with only a single σ_- ($\Delta m_J = -1$), σ_+ ($\Delta m_J = +1$), and π ($\Delta m_J = 0$) transition. To further simplify the measured fluorescence, only the σ transitions were excited by an appropriate choice of the laser polarization direction with respect to the external magnetic field direction. As discussed above, with a sufficiently broad laser this splitting will result in a quantum beat frequency proportional to the frequency separation between the two σ energy levels, and therefore is proportional to the magnetic field. Thus, if the quantum beat frequency is measured and the Landé-g factor for the unsplit state is known, the magnetic field strength is directly computed:

$$B = \frac{\omega \hbar}{2\beta g_J}. \quad (4)$$

The Landé-g factor for the $J=1$ state of the chosen neutral argon transition has been measured to be 1.38003 ± 0.00040 [26]. For the $J=1$ state of the neutral helium transition, the theoretical Landé-g factor based on LS-coupling is 1.

3. Experimental configuration

3.1. Experimental facility

Figure 3 presents a simplified block diagram of the overall experimental configuration. Neutral argon and helium is investigated in the Compact Helicon for Waves and Instabilities Experiment (CHEWIE) plasma device. The CHEWIE facility has undergone several iterations and is described as it was configured during these experiments. CHEWIE is a horizontally oriented helicon plasma device that serves as a test stand for the PHase Space Mapping facility [27]. The central chamber in CHEWIE is an $\varnothing 8$ inch 6-way stainless steel cross. A base pressure of 10^{-8} Torr is maintained with a TSH 522 turbo molecular pumping station at the bottom of the six-way cross. Plasma is created at 13.56 MHz with 50 W (argon) and 500 W (helium) of forward power through a copper strap antenna wrapped around a $\varnothing 3$ inch, 24 inch long PyrexTM tube mounted on one side of the 6-way cross. The gas flow into the vacuum vessel is controlled with a MKS mass flow controller with a maximum throughput of 30 SCCM. The gas pressure is monitored with a KJL300800 full range pressure gauge. Four water-cooled electromagnets, connected in series, surround the vacuum vessel, providing a guide magnetic field along the mechanical axis of the device between 0 and ~ 730 G. The resolution of the magnetic field

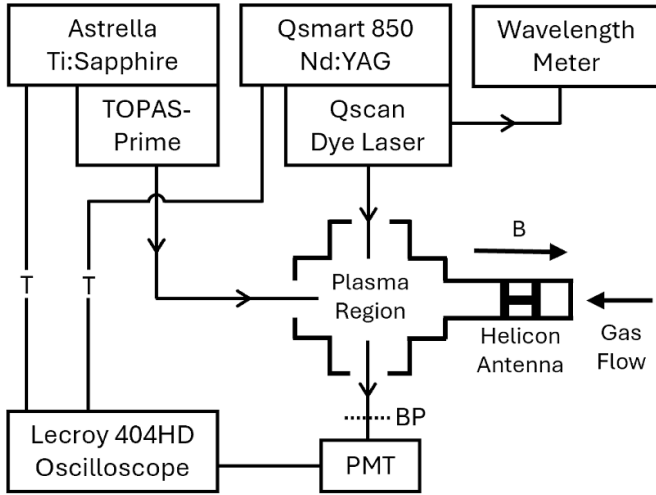


Figure 3. Simplified experimental configuration diagram. The lasers have polarized outputs; laser polarizations in the plasma region are perpendicular to B . T = trigger from laser to oscilloscope, BP = bandpass filter, PMT = Hamamatsu 11526-20-NF photomultiplier tube.

control amperage is 1 A, corresponding to an uncertainty of 2.44 G at the measurement location.

3.2. Laser systems, injection, and collection

Two different lasers of varying spectral bandwidths were employed in this work. For measurements probing field strengths up to ~ 70 G, a Quantel Qscan pulsed dye laser system has been used. For measurements probing field strengths as large as ~ 320 G, an ultra-short fs pulsed laser has been used. Care is taken when designing the beam paths to ensure the polarization of the lasers are roughly perpendicular to the magnetic field in the collection region.

The Quantel Qscan includes a Quantel Qsmart 850 pulsed neodymium-doped yttrium–aluminum–garnet (Nd:YAG) laser that generates >850 mJ of light at 1064 nm over ~ 6 ns pulses at a 10 Hz cadence. These pulses are frequency doubled to create >430 mJ of 532 nm light. This light ‘pumps’ a Styryl 11 dye solution in a series of dye cells to generate ~ 25 mJ of 772.6333 nm (vac.) light, with a spectral width of ~ 1 GHz (2 pm). Before the laser exit, a portion of the laser beam (approximately 15%) is steered into a single-mode fiber coupled to an Angstrom UVII WS6-200 wavelength meter for real time monitoring of the laser wavelength. The remainder of the beam is delivered to the vacuum vessel by a set of simple dielectric mirrors optimized for wavelengths between 750 and 1050 nm. The light is steered to inject radially into CHEWIE, carefully maintaining the vertical polarization of the beam such that the polarization is perpendicular to the magnetic field. This ensures that only σ -transitions are excited.

The fs pulsed laser system is a Coherent Astrella ultra-short pulsed Ti:Sapphire solid-state laser system equipped with a Light Conversion TOPAS-Prime optical parametric amplifier and frequency harmonic units. This laser is capable of generating ~ 100 fs pulses of light between 190 and 2600 nm at a

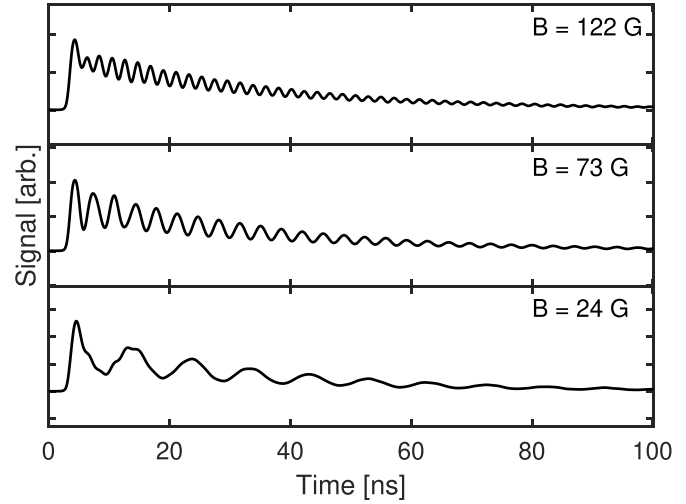


Figure 4. Samples of fluorescence time series illustrating quantum beat oscillations for select magnetic field strengths measured using the fs laser system.

1 kHz cadence. Approximately 188 μJ of 772 nm light (argon) and 65 μJ of 501 nm light (helium) is produced by this laser system. Due to the ultra-short pulse width of this laser, the spectral width of this laser is ~ 7500 GHz (15 nm), significantly larger than the pulsed ns laser system. Therefore, this system can probe much stronger magnetic fields. Since the spectral width is so large, it is not necessary to actively monitor the output wavelength of this laser system. The fs pulses are injected axially into CHEWIE. The axial injection direction guarantees the polarization is perpendicular to the magnetic field, again ensuring that only σ -transitions are excited.

Fluorescence is collected perpendicular to both laser injection directions and to the magnetic field direction using a simple set of $\varnothing 2$ inch lenses. The collected light is left unpolarized. The collected light is focused into a $\varnothing 200$ μm core, 0.22 NA multimode fiber optic cable coupled to a Hamamatsu H11526-20-NF photomultiplier tube (PMT) that amplifies the collection fluorescence. Background plasma emission and ambient radiation is rejected by a bandpass filter centered around the fluorescence wavelength with a full-width at half maximum of 1 nm for the neutral argon measurements. No emission filtering was used for the neutral helium measurements. The filtered amplified signal is digitized with a Lecroy 20 GS s^{-1} , 4 GHz, multi-channel WavePro 404HD oscilloscope.

The collection cone spot size is ~ 200 μm , while the lasers were collimated at ~ 3 mm diameter. These overlapping regions give a spatial resolution of <1 mm^3 .

4. Results and discussion

The fluorescent emission lifetime of the chosen Ar I transition was measured to be ~ 100 ns as shown in figure 4. In order to exclude any scattered light from the laser, the first 20 ns of the PMT time series were discarded for measurements using both the ns and fs laser systems. The standard exponential

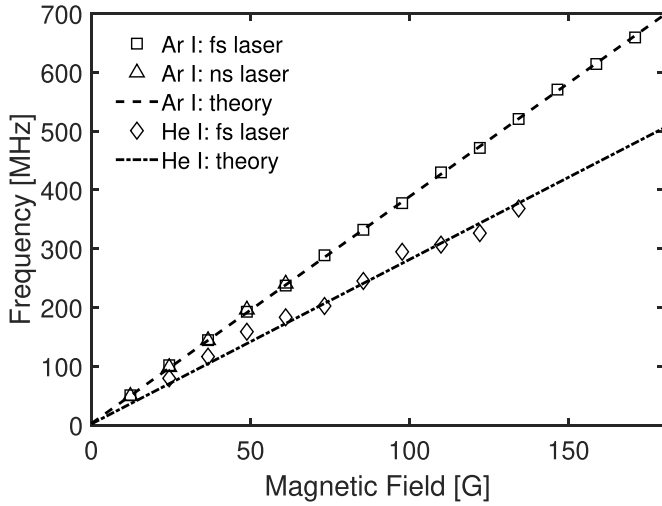


Figure 5. Applied magnetic field versus the measured peak FFT frequency for neutral argon using the ns and fs lasers, and for neutral helium using the fs laser. Magnetic fields values have a precision of ± 1.2 G.

fluorescent decay was subtracted off and the beat frequency calculated by fast-Fourier transform (FFT). Figure 4 shows examples of the beat frequency increasing as the magnetic field increases. The amplitude of the oscillations decreases exponentially in time as predicted by theory and also with increasing frequency.

During measurements using the Qscan ns laser system in neutral argon, the magnetic field magnitude was varied from 0 to 85 G (with an uncertainty of ± 1.2 G) and 100 ns time series of fluorescent emission recorded at each discrete field strength. Frequencies were obtained through FFT of each time series with a resolution of ~ 13 MHz. The precision of the recorded frequency was improved to 0.1 MHz through zero-padding. For field strengths above 73 G (282 MHz σ -state separation), there was no detectable quantum beat signature observed in the fluorescence. Even though the bandwidth of the laser is comparable to the σ -state separation at that field strength, the laser intensity at the edges of the laser line shape is too small to yield a detectable QBS signal. The relation between the magnetic field and the measured beat frequency is shown in figure 5. A linear least-squares fit was performed to determine the Landé-g factor. A background field of ~ 0.7 G was measured with a calibrated Gaussmeter and that value was used as the fit intercept. The fitted slope was 3.95 ± 0.09 MHz G $^{-1}$, which corresponds to a Landé-g factor of 1.41 ± 0.03 , a 2% difference from the literature value of 1.38.

Changing the laser wavelength from the rest frame absorption wavelength results in a drop in the overall fluorescence amplitude, with the amplitude following the neutral velocity distribution function (VDF) as with pulsed LIF [28]. For different pump wavelengths, the relative amplitude of quantum beat oscillation compared to the total fluorescent amplitude remains approximately constant, with the beat amplitude also following the neutral VDF (figure 6).

The beat signatures observed in these measurements exhibit a strong sensitivity to the laser energy. Normalizing the beat

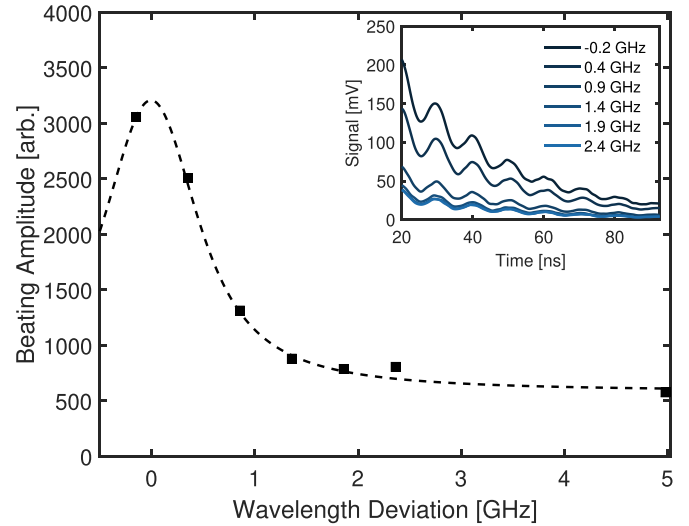


Figure 6. Measured beat frequency amplitude (for a magnetic field strength of 24 G) versus the laser wavelength's deviation from the predicted absorption wavelength. Inset depicts PMT trace for each data point from the predicted absorption wavelength (darkest blue) to 2.4 GHz deviation (lightest blue).

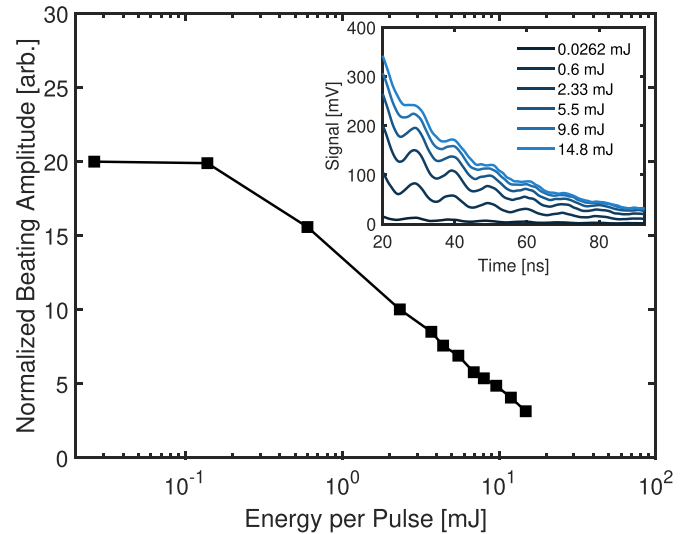


Figure 7. Amplitude of the beat frequency normalized to the total measured fluorescence at varying injection energies for a magnetic field strength of 24 G. Shown in the inset are measured fluorescence time traces shown for sampling of energies between $26 \mu\text{J pl}^{-1}$ (darkest blue) to 14.8 mJ pl^{-1} (lightest blue).

amplitude to the amplitude of the fluorescence at 20 ns, we find a steady decrease as laser energy increases from $138 \mu\text{J pl}^{-1}$ to 14.8 mJ pl^{-1} (figure 7). The normalized beat amplitude levels out at lower laser pulse energies to a maximum relative amplitude. Although the beat amplitude relative to the overall fluorescence increases with lower energy, there is clearly an optimal balance between the signal-to-noise of the beat in the time series and the overall signal amplitude.

We hypothesize that this effect results from stimulated emission for larger pulse energies (akin to what causes power

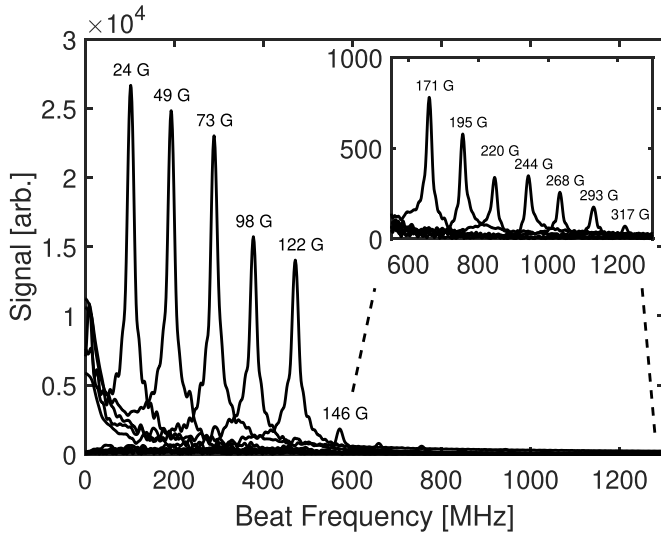


Figure 8. Examples of power spectra of the fluorescence time series for varying magnetic fields and using the fs laser.

saturation when using high power pulses in LIF). As incoming photons with a wavelength matching one of the possible transitions stimulate emission, they act as a measurement for which upper state the electron is in, removing the uncertainty and therefore the beating. In other words, the stimulated decay photon is identical to the pumping photon and therefore no interference effects occur. This is an important effect that limits the laser pulse energy to be used in QBS even if larger pulse energies yield larger fluorescence amplitudes. For this technique, more is not ‘better.’ This effect may limit the laser focal spot size and thus the spatial resolution for a given laser energy.

To extend the dynamic range of this diagnostic, a larger laser linewidth is needed. The fs Astrella laser system was used to measure magnetic field strengths as large as 317 G, corresponding to a state splitting of 1225 MHz. Figure 8 shows a sampling of power spectra of the fluorescent time series for discrete applied magnetic fields ranging from 24 G to 317 G. As the magnetic field and hence the beat frequency increases, the amplitude of the oscillation steadily decreases. There is a rapid decrease in the beat amplitude around 500 MHz (roughly corresponding to the expected bandwidth of the PMT), but peaks are still discernible up to ~ 1.2 GHz. The relationship between the beat frequency and the magnetic field is shown in figure 5, with the linear least-squares line of fit giving a slope of 3.86 ± 0.01 MHz G $^{-1}$ (the intercept once again fixed by the known background of 0.7 G). This slope corresponds to a Landé- g factor of 1.379 ± 0.005 , a 0.1% deviation from the literature value. For measurements recorded using both laser systems, a minimum detectable field of 4.8 G (~ 18.7 MHz beat frequency) was achieved, limited by the electromagnet power supply step size and precision and by the lifetime of the electron transition scheme used. An estimated minimum detectable field strength of ~ 3.2 G (~ 12.5 MHz beat frequency) is estimated based on the fluorescence decay time used here of

~ 80 ns. Using the full lifetime of 100 ns would extend this to ~ 2.6 G (~ 10 MHz beat frequency).

Preliminary measurements were performed in neutral helium, with the fluorescence emission unfiltered. The magnetic field was varied from 12 to 244 G. Multiple fluorescent lifetimes were observed simultaneously due to the lack of bandpass filter in the collection path and the overall lifetime for the quantum beat decay was observed to be ~ 40 ns. The correlation between magnetic field and beat frequency is shown in figure 5. Multiple high frequency signals present in the unfiltered fluorescence skewed higher magnetic field measurements, thus only measurements up through 146 G are used. A fixed intercept of 0.7 G is once again used, with a linear least-squares line of fit giving a slope of 2.8 ± 0.1 MHz G $^{-1}$. This corresponds to a Landé- g factor of 1.00 ± 0.04 , which matches exactly with the theoretical value calculated from LS-coupling theory.

A statistical analysis of the magnetic field was performed based on 2000 individual laser pulses. In figure 9 we show how the mean and standard deviation of the magnetic field depend on the number of individual laser pulse time series averaged. Figure 9(a) shows an example of the statistical mean magnetic field magnitude derived as a function of laser pulses for an applied magnetic field strength of 183 G and injected laser energy of 46 μ J in an argon plasma. After averaging as few as only four laser pulses, the magnitude of the applied field is confidently derived with an uncertainty of less than ± 0.8 G, making this technique ideal to accurately and efficiently detect transient electromagnetic events in plasma environments. Sub-Gauss uncertainty is achieved in fewer averaged pulses (reaching single-shot acquisition) with higher laser energies and/or with lower magnetic fields as seen in figure 9(b) due to the increased signal-to-noise and relative beat amplitude respectively.

There are three principal limitations of this diagnostic. A $J=0$ metastable population with an allowed $J=1$ transition is preferred to easily resolve the quantum beating due to Zeeman splitting. With more complicated initial states, the additional beat frequencies from more transitions may obscure each other. The lifetime of the state decay sets a lower limit on the resolvable frequency, as well as on the frequency resolution, and therefore a limit on the minimum measurable magnetic field. The limitation suggests that highly collisional systems with rapidly quenched metastable populations and short upper state lifetimes may prove sub-optimal for this diagnostic. Finally, the bandwidth and sampling rates of the detection electronics, such as the PMT and oscilloscope, as well as the linewidth of the laser, set an upper limit on the resolvable frequency.

For an argon plasma, the magnetic field was reliably measured up to 317 G with a femtosecond laser, and up to ~ 63 G with a nanosecond laser. The upper limit on our femtosecond laser measurement is currently the response time of our PMT. This upper limit could be increased on the order of $\sim 10^3$ – 10^4 G by upgrading to a hybrid micro-channel plate/PMT or a streak camera and higher bandwidth cabling and electronics. This upper limit would increase for lower Landé- g factor

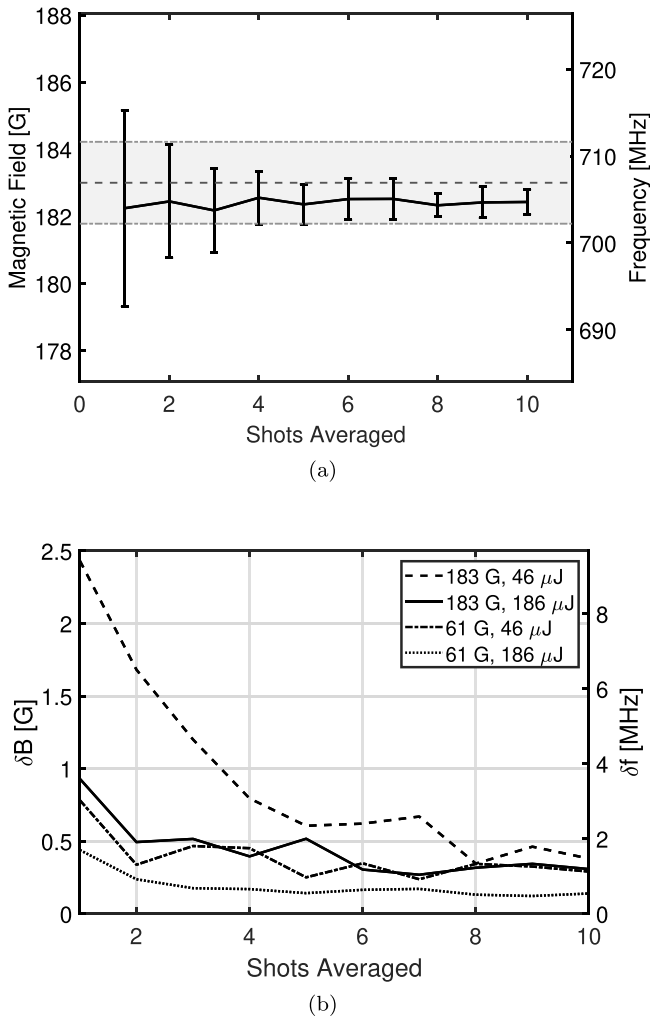


Figure 9. (a) The mean of the magnetic field measurements (left axis) and associated mean of the beat frequency (right axis) for a 183 G magnetic field with 46 μ J energy per pulse. The dot-dash bounded grey region indicates the expected magnetic field of 183 ± 1.2 G. (b) A sampling of the standard deviation of magnetic field measurements for varying magnetic fields and laser energies.

upper states like the helium line investigated, and decrease for higher Landé-g factor upper states. A picosecond laser may prove optimal for providing a spectral width broad enough for large magnetic field measurements, but narrow enough to minimize excitation of other nearby electron transitions. The maximum lifetime of the chosen neutral argon transition (~ 100 ns) limits the frequency splitting resolvable by FFT to ~ 10 MHz (2.6 G), or ~ 25 MHz (8.9 G) for the neutral helium transition (~ 40 ns lifetime). These lifetimes are partially dependent on the system characteristics and may be shortened by issues such as high collisionality. The decision to ignore the first 20 ns of the PMT was specific to the nature of the scattered laser light in our system, and will vary in other systems. If the scattered light from the laser pulse is well characterized, it can be subtracted

from the PMT trace to increase the usable fluorescent lifetime. The size of these frequency bins can be decreased with zero-padding of the FFT to improve the precision of the measurement. Any frequencies within these bins near the quantum beat frequency will still be indistinguishable and may introduce a small shift in the magnetic field measurement. No splitting of less than this minimum bin size can be accurately measured, preventing measurements of fields < 2.6 G (argon) and < 8.9 G (helium). Zero-padding was also observed to introduce a small shift in the peak frequency at frequencies after the cutoff seen with the fs laser, slowly growing to a deviation of ~ 8 MHz at frequencies over 1 GHz.

5. Conclusion

Here we have presented an exploration of the application of QBS to a low pressure neutral argon plasma as a diagnostic to measure magnetic field strengths of up to 317 G with a resolution of < 1 G. Using a laser with a broad laser bandwidth (i.e. a pulsed laser), both σ transitions are excited so long as the separation between the $m_J = +1$ and $m_J = -1$ Zeeman states of the upper electron energy level is kept small relative to the laser bandwidth. The narrower bandwidth (~ 1 GHz) nanosecond laser was limited to measuring up to 281 MHz state splitting, while the broader bandwidth femtosecond laser measurements were limited by the bandwidth of the detection electronics.

This work has demonstrated QBS to be a spatially and temporally resolved diagnostic for magnetic fields within argon and helium plasmas within low-pressure laboratory conditions. The spatial resolution of this technique (determined by the overlapping laser injection and fluorescence collection regions) was shown to be restricted by the laser energy due to the reduction in beat amplitude, though this relation was not thoroughly investigated. A spatial resolution of < 1 mm³ and a temporal resolution of ~ 100 ns were achieved.

Care was taken to eliminate π -transitions, maximizing the beating signal at the 2σ -frequency. Changing the polarization to include both π - and σ -transitions should introduce an additional beat frequency at half the 2σ -frequency investigated in this work, with the tradeoff of lower signal at each frequency. This may prove useful for achieving higher magnetic field strength measurements with lower bandwidth electronics and is worth investigation.

While favorable electron transition schemes for neutral argon and helium were identified and proven in this work, this diagnostic should be applicable to other species with $J = 0$ metastable populations.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgments

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